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Influence of essential oils supplementation on rumen fermentation profile and ruminal microbial population *in vitro*

Prem Prashant Chaudhary^{1*#}, Navneet Goel¹, Greg Baker², Jyoti Saxena³, Nasib Singh⁴, Indu Chaturvedi⁵, Ashwani Sharma⁵ and Sunil Kumar Sirohi¹

¹Nutrition Biotechnology Laboratory, Dairy Cattle Nutrition Division, National Dairy Research Institute, Karnal (India), [#]Current address: Department of Internal Medicine, University of Michigan, Ann Arbor, Michigan, Email: luckyppgi@gmail.com, Phone: +1-734-780-0500

²School of Pharmacy and Life Sciences, The Robert Gordon University, Garthdee Road, Aberdeen (Scotland)

³Biochemical Engineering Department, BCT Kumaon Engineering College, Dwarahat, Uttarakhand (India)

⁴Department of Biotechnology, Eternal University, Baru Sahib- 173101, Himachal Pradesh (India)

⁵Indian Science and Technology Foundation (ISTF), C-1/31, Yamuna Vihar, New Delhi-110053

* Corresponding author

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ABSTRACT

In this study, we evaluated the effect of plant essential oils on methane production and ruminal fermentation parameters. Five essential oils were initially screened for their efficacy to decrease methane production and enhance the digestibility of wheat straw (roughage: concentrate 50:50) based diets. On the basis of preliminary results, organum and thyme oils were further investigated with three variable combinations of wheat straw and roughage: concentrate ratio. These oils significantly decreased the methane production whereas the digestibility and total volatile fatty acids level in most of the treatments varies differentially. Individual volatile fatty acid (IVFA's) analysis indicated that acetate percentage was decreased in high fiber diet (HFD), but showed an increasing pattern in medium fiber diet (MFD) and low fiber diet (LFD) following addition of organum and thyme oils. Propionate levels exhibited decline in majority of the treatment combinations. However, butyrate levels witnessed significant increase compared to control diets. NH₃:N₂ ratio which varies up and down with the treatment with both the oils. qRT-PCR analysis revealed significant reduction in methanogen and fungal counts following addition of thyme oil whereas an increase in methanogen counts and decrease in fungal counts was observed with organum oil supplementation. Further, total bacterial population remains unchanged but protozoal population revealed increase after oil supplementation. These findings indicated the significant influence of organum and thyme oil on rumen fermentation mediated through modulation of microbial population.

INTRODUCTION

There is an urgent need for improvement of feed utilization in dairy animals by modulating the ruminal microbiome to achieve the dual goals of reducing the methane (CH₄) emissions and increasing the milk yield as well as the availability of good quality meat (1-4). In the recent years, several bioactive plant

compounds such as essential oils (EOs), saponins, tannins, and flavonoids have been evaluated for their ability to improve feed utilization efficiency by modulating rumen microbial fermentation processes (5-7). These bioactive compounds are considered as the first choice for rumen manipulation owing to

their natural origin. Further, in the context of global warming, reducing CH₄ emission from the ruminants as the methane emitted from the ruminants contribute majorly to the global warming (8-9). Therefore, reducing CH₄ emissions from ruminants can substantially reduce greenhouse gases and increase the efficiency of livestock production. Different strategies based on chemical inhibitors, essential oils, ionophores etc. have been explored to reduce CH₄ from the ruminants without affecting other nutritional parameters (10). Moreover, several molecular diversity studies on the rumen methanogens were also performed to determine the dominance of methanogen genera or orders so that strategies could be designed to suppress methanogens population with ultimate aim of mitigating CH₄ emission (11-14).

Out of the various plant bioactive compounds, EOs are usually preferred as these are found to be beneficial for ruminal microbial metabolism. EOs can reduce the risk of rumen acidosis, decreasing intra ruminal nitrogen turnover and nitrogen excretion and also inhibits methanogenesis (9). Supplementation of EOs was also found to increase milk yield (15). However, detrimental effects of EOs supplementation on rumen fermentation and fibre digestion have also been reported (16-17). Therefore, it is worthwhile to validate the effects of EOs on rumen fermentation patterns, CH₄ emission and rumen microbial communities to exploit EOs for in field applications. The present study was performed with an aim to check the dose dependent effect of EOs on rumen fermentation pattern and population of rumen methanogenic archaea, total rumen bacteria, protozoa and fungi counts *in vitro*.

2. Material and methods

2.1. Essential oils (EOs)

Five EOs i.e., cinnamon leaf oil, clove bud oil, thyme oil, origanum oil and rosemary oil were purchased from Sigma-Aldrich (Sigma-Aldrich, St. Louis, MO) and used for the present study. EOs was tested at 3 doses viz. 0.05 g/L, 0.15 g/L, and 0.45 g/L. Experimental controls without EOs were included in various experimental setups.

2.2. Rumen liquor collection and *in vitro* fermentation

The total mixed ration (TMR) of the buffaloes was composed of roughage concentrate ratio of 50:50. Chemical composition of TMR is provided in Table 1. The buffaloes were fed the TMR (18.5 kg) twice a day at 8 AM and 6 PM. Inocula for *in vitro* incubation were collected from 3 fistulated male Murrah buffaloes. About 1000 ml of rumen sample was collected into a 1000 ml bottle, leaving no headspace in the sample bottles. The samples were brought to the laboratory within 10 min (about 1 km away) and then placed into an anaerobic chamber containing N₂ (95%), H₂ (3%), and CO₂ (2%). Equal volumes of the rumen fluid collected from each of the buffaloes were combined as the inoculum. *In vitro* incubations were carried out in 120 ml serum bottles in triplicate. Medium fibre diet (MFD) having roughage and concentrate, in a ratio of 50:50 was used as the substrate for screening studies. Three different diets viz. high fibre diet (HFD), MFD and low fibre diet (LFD) were used (Table 1). *In vitro* buffered medium was prepared anaerobically and Hohenheim gas test apparatus was used as described previously (18). Three sets of syringes were prepared in triplicate using three different diets (200 mg) as substrate in syringes. Buffered medium (30 ml) containing rumen liquor was dispensed into the syringes and incubated at 39°C for 24 h with occasional manual shaking.

2.3. *In vitro* analyses of fermentation gases and volatile fatty acids (VFAs)

At the end of 24 h of incubation, gas pressure in the culture bottles was measured using a manometer (Traceable; Fisher Scientific) to determine total gas production. Then, 10 ml of headspace gas was collected into a tube filled with distilled water by displacement. The liquid samples (1 ml) were individually collected into microcentrifuge tubes and preserved at 20°C for microbial analysis. The remaining culture volume was filtered through filter bags (Ankom Technology) to determine the degradability of the added substrate. The filtrates were sampled in microcentrifuge tubes for VFAs and NH₃ measurement. If not analyzed immediately, all the samples were stored at 20°C until further processing. Concentrations of CH₄ in the gas samples were determined using a gas chromatograph (Nucon-5765, India) as described previously (19-21). VFA concentrations in the cultures were also analyzed by a gas chromatograph (GC; Nucon-5765) equipped with flame ionization detector and stainless steel column packed with Porapak-Q (length 6'; o.d. 1/8" i.d. 2 mm; mesh range 80–100; (21). The concentrations of NH₃ in the fermentation cultures were measured by a calorimetric method (22). The apparent DM degradability of the substrate was determined after drying the residues collected in the fiber bags and the initial substrate at 105°C in a hot air oven for 24 h.

2.4. DNA extraction

DNA was extracted from 0.5 ml of homogenized sample using the repeated bead beating (on a beadbeater; BioSpec Products, Bartlesville, OK) and column purification by using DNA extraction kit (Fermentas, USA). DNA quality was evaluated using agarose gel (1%) electrophoresis (23), and DNA

concentrations were measured in NanoQuant instrument (Tecan, USA). Samples were stored at 20°C until further analysis.

2.5. Quantitative real-time PCR (qPCR) measurements

The population sizes of total bacteria, total methanogens, total fungi and total protozoa were quantified with SYBR-based qPCR in MJ Mini Opticon Real-Time PCR System (Bio-Rad, USA) using specific primers (Table 2) as described previously (27). To minimize potential bias, instead of a single strain, sample-derived qPCR standards were prepared using the respective specific PCR primer set and a composite metagenomic DNA sample that were prepared by pooling equal amounts of all the metagenomic DNA samples (14, 28). Amplification of the respective target genes was performed and cloned into pGEM-T Easy vector (Promega, Madison, WI). The purified plasmids were quantified by Nano-drop spectrophotometer (Thermo Fischer, USA) with multiple dilutions. Target DNA was quantified by using 10-fold serial dilutions from 10⁻¹ to 10⁻⁸ plasmid copies of the previously quantified plasmid standards. To eliminate the effect from potential primer dimers in the SYBR-based qPCR assays, the fluorescence signal was acquired at 86°C, at which primer dimers were completely denatured and thus not detected, and used in quantifying populations of the microbial groups. The qPCR assay for each group was performed in triplicate. The absolute abundance was expressed as number of gene copies/ 50 ng of DNA.

2.6. Statistical analysis

Experimental data of different parameters was analyzed in factorial design by using OPSTAT statistical software developed by Chaudhary Charan Singh Haryana Agriculture University, Hisar, India (29). Data analysis for detailed set

of EOs was analyzed in $2 \times 3 \times 3$ factorial design. When the overall *F*-test was significant, differences between means and the control were declared significant at $P \leq 0.05$ using the Fisher's Least-significant-difference.

3. Results and discussion

3.1. Screening of EOs for methane production and digestibility *in vitro*

Five EOs i.e. cinnamon leaf oil, clove bud oil, thyme oil, origanum oil and rosemary oil were studied for their efficacy towards methane reduction and digestibility in wheat straw based diets. Cinnamon oil at a dose of 450 mg/l was found to be effective against methanogens and decreased CH_4 production slightly as compared to the control samples (Table 3). This decrease in CH_4 production is might be due to eugenol, an active component in cinnamon oil (accounting up to 8%) which has antimicrobial action against bacterial populations involved in rumen fermentation [7]. However, clove bud oil was failed to exhibit CH_4 reduction at all dose levels. In case of thyme and origanum oil, significant CH_4 reduction (68.8 and 82.5%, respectively) was observed at 450 mg/L dose (Table 3). The inhibitory effect of these two oils at higher dose was statistically significant to ($p \leq 0.05$). These findings are in agreement with previous studies where thymol, a main component of EO derived from *Thymus* and *Origanum* plants, was a strong inhibitor of CH_4 *in vitro* (30). Similarly, Macheboeuf et al. (17) reported suppression of CH_4 to the extent of 99% at 6 mM dose of *Origanum vulgare* EO and its component, thymol.

Digestibility patterns remain unaltered by the addition of cinnamon EO. These findings are in contrast to an earlier study where decrease in the digestibility of dry matter (DM) was observed at higher dose of cinnamon leaf oil (31). Clove bud oil, on the

other hand, increased the digestibility in a dose dependent manner. Thyme oil reduced the digestibility at higher dose whereas an improvement in digestibility was observed at lower doses. There was no significant shift in DM digestibility in origanum oil treated samples as compared to that of control at 50 mg/L and 450 mg/L concentrations. Rosemary oil was ineffective in either decreasing CH_4 production or enhancing the DM digestibility (Table 3). On the basis of the screening results, thyme and origanum oils were selected for further detailed study in different diets formulated using wheat straw in different ratio of roughage and concentrate i.e. 60:40, 50:50 and 40:60 to study the effect of supplementation of these compound and oils on CH_4 production, digestibility and other fermentation parameters.

3.2. Effect of EOs supplementation in wheat straw based diet combinations

3.2.1. Effect of EOs on CH_4 production

When origanum oil and thyme oil were supplemented at 450 mg/l dose in HFD having 60:40 roughage: concentrate ratio, CH_4 production was reduced up to 14.42 and 12.13 ml/g of substrate as compared to 23.46 ml/g of substrate ($p \leq 0.05$) in untreated control diet. Similar pattern of reduction in CH_4 levels was observed in terms of mM/g of DM or per g DDM in MFD and LFD (Table 4). These findings indicated that CH_4 production was significantly decreased (50-80%) by addition of thyme and origanum oil in wheat straw based diets at 450 mg/L concentration. In some previous studies, Crane et al. (32) described the inhibitory effect of limonene on CH_4 formation whereas Macheboeuf et al. (17) found that EOs extracts from *Origanum vulgare* (20% and 35% thymol) and *Thymus vulgare* (20% p-cymene) significantly reduced CH_4 production at dose of <2 mM [12, 22].

Wang et al.(33) showed that inclusion of 0.25 g/day of EOs mixture from oregano plants in the diet of sheep for 15 days lowered CH₄, while Beauchemin and McGinn (34) found no significant effect of EOs on methanogenesis when fed to beef cattle (1 g/day) for 21 days. In another study, Sallam et al. (35) found EO from *Mentha microphylla* to be an effective methane mitigating agent at 25 µl/75 ml dose levels under in vitro fermentation. Overall, it can be concluded that the effect of EOs on methanogenesis is diet and dose dependent.

3.2.2. Effect of EOson *in vitro* dry matter digestibility (IVDMD)

IVDMD was decreased from 76.19% in case of control diet to 74.78% and 56.87% ($p \leq 0.05$). in origanum and thyme oil treatments, respectively when evaluated in HFD (Table 4). In case of MFD, IVDMD was again significantly ($p \leq 0.05$) reduced to 60.66 and 57.57% when origanum and thyme oil, respectively was supplemented as compared to 77.07% in control diet. Similar effect was evident in case of LFD group. IVDMD was significantly decreased in all three diet regimen except in case of origanum oil treatment with HFD. These results are in agreement with Benchaar et al. (36) findings in which thymol, origanum oil and carvacrol manifested decrease in the digestibility. One possible explanation for this decrease in IVDMD may be the effects of EOs on rumen microbes which are involved in fibre degradation.

3.2.3. Effect of EOson Total Volatile Fatty Acid (TVFA) and Individual Volatile Fatty Acids (IVFA)

TVFA levels exhibited decline when supplemented with origanum ($p \leq 0.05$) and thyme oil as compared to control diet (Table 5). In MFD on supplementation of origanum oil, TVFA was increased to 9.08 mM/100 ml

as compared to 8.17 mM/100 ml in control diet. In contrast, TVFA production was decreased significantly ($p \leq 0.05$) on thyme oil supplementation up to 7.00 mM/100 ml as compared to 8.17 mM/100 ml in control diet. In case of LFD also, TVFA concentrations were significantly ($p \leq 0.05$) decreased (Table 5). However, the effects of origanum and thyme oil were not statistically significant. Castillejos et al. (37) reported that lavandin and oregano at 500 mg/l dose inhibited rumen microbial fermentation thereby decreasing VFA concentration [8]. This report is in agreement with our findings where decrease in the TVFA concentration was observed at 450 mg/L doses of thyme and origanum oil. However, Benchaaret al. (36) reported increase in the concentration of TVFA on supplementation of thymol when supplemented at lower doses.

Propionate production was decreased ($p \leq 0.05$) whereas butyrate production was increased ($p \leq 0.05$) when origanum oil and thyme oil were supplemented in HFD, respectively. However, decline was witnessed in acetate levels as shown in Table 5. In case of MFD, acetate and butyrate production was found to be increased ($p \leq 0.05$) whereas propionate levels decreased ($p \leq 0.05$). Similar trend was observed in case of LFD. These observations are in agreement with Benchaar et al. (34) in which addition of carvacrol and thymol resulted in increased butyrate levels ($p < 0.05$) and lowered levels of propionate but no change in acetate. EOs at 450 mg/l dose level may be affecting the fibre degrading microbes as evident by reduced IVDMD and acetate percentage which may affect the propionate production too.

3.2.4. Effect of EOs on ammonia nitrogen

In HFD group, ammonia nitrogen was decreased upto 28.47 mg/ 100 ml and 15.4

mg/100 ml in origanum oil and thyme oil ($p \leq 0.05$) treated samples, respectively as compared to 39.2 mg/100 ml in control diet (Table 5). In MFD and LFD groups, ammonia nitrogen was increased in case of origanum oil whereas decreased with thyme oil treatment (Table 5). Thyme oil supplementation might be diverting some energy or affecting particular group of rumen microbes which in turn increase microbial protein synthesis. MBM yield was increased significantly in majority of the groups treated with thyme oil. Similarly, the increased PF values further support our findings. Borchers (38) found that the addition of thymol to rumen fluid *in vitro* resulted in the accumulation of AA N and the reduction of ammonia nitrogen concentrations suggesting that thymol inhibited deamination.

3.3. Effect of EOs on methanogens and other rumen microbes

Real time quantification of methanogens and other rumen microbes in origanum and thyme oil supplemented wheat straw based diet is shown in Table 6. Copy number of total bacteria was 2.12×10^{10} , 2.71×10^{10} and 1.98×10^{10} per 50 ng of DNA in control, origanum and thyme oil treated samples, respectively. Apparently, the total bacterial population was lowest in case of thyme oil and highest in case of origanum oil. However, the differences in copy number values were statistically similar. This fact is also supported by a recent study where supplementation of *Eucalyptus* oil was ineffective against viable bacteria during rumen fermentation (38). Methanogens population witnessed decrease in thyme oil-treated group (5.6×10^7) whereas significant increase was noticed with origanum oil supplementation (6.7×10^7 ; $p \leq 0.05$) as compared to control diet (5.9×10^7 per 50 ng of DNA). Protozoal population exhibited significant increase following supplementation of origanum and thyme oils in wheat straw

based diet in comparison to control diet. These findings are, however in contrast with a recent study where protozoa populations were reduced by supplementation of *Eucalyptus* crude oil (39). This might be explained on the basis of the activity of active components present in different EOs. However, the total fungal population was decreased significantly in origanum and thyme oil supplementation samples than control diet. Copy number of total fungi population in control, origanum and thyme oil treated diets was 9.6×10^7 , 8.9×10^7 and 7.8×10^7 , respectively. Decrease in fungal counts was in accordance to study of Talebzadeh and Alipour (40) where decrease in *Neocalimastix* growth was observed on addition of ajowan EOs. Other works also done on methane inhibition (41-45). These two oils directly affected the methanogens population which was increased in case of origanum oil but significantly decreased by thyme oil. Further, the total fungi population was significantly decreased, protozoal counts were significantly increased on supplementation of both of these oils whereas total bacterial population was not affected much by supplementation.

4. Conclusion

Origanum and thyme supplementation resulted in modulation of rumen fermentation parameters as well manifested effects on the populations of ruminal microbial groups. However, a conclusive relationship between CH_4 reduction and dynamics of major ruminal microbial groups was not directly evident. These findings necessitated further evaluations of essential oils at a broad range of doses or various diets combinations to utilize these plant metabolites for effective methane mitigation.

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Disclosure statement

The authors declare that they have no competing interests.

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Table 1. Chemical composition of different diets (percentage on DM basis) used for experimental analyses.

Diets	Chemical (g/kg on constituents of diets DM basis)		
	D1 (60:40)	D2 (50:50)	D3 (40:60)
OM	86.76	87.84	87.56
CP	10.86	12.53	14.27
EE	2.34	3.04	3.48
NDF	62.31	60.45	53.87
ADF	37.20	32.95	29.87
HC	25.11	27.50	24.00
Cellulose	25.66	21.80	19.90
ADL	5.10	5.06	5.36
Total ash	13.24	12.16	12.44
Starch	11.47	13.18	17.44

OM= organic matter, CP= crude protein, EE=Ether extract , NDF=Neutral Detergent Fiber, ADF= Acid Detergent Fiber, HC= Hemicelluloses , ADL= Acid Detergent Lignin

Table 2. Details of the primers used for Real Time-PCR estimation of rumen microbial groups.

Primer pair	Primer sequences (5' – 3')	Amplicon length (bp)	Reference
<i>mcr-A</i> (Total methanogens)	Fwd 5'-TTCGGTGGATCD CARAGRGC-3' Rev 5'-GBARGTCGWAWCCGTAGAATCC-3'	145	(24)
16S rRNA (Total bacteria)	Fwd 5'-CGGCAACGAGCGCAACCC-3' Rev 5'-CCATTGTAGCACGTGTGTAGCC-3'	160	(25)
18S rRNA (Total Protozoa)	Fwd 5'-GAGCTA ATACATGCTAAGGC -3' Rev 5'-CCCTCACTACAATCGAGATTTAAGG-3'	321	(26)
Internal Transcribed Spaces (ITS)	Fwd 5'-GAGGAAGTAAAAGTCGTAAACAAGGTTT-3' Rev 5'-CAAATTCACAAAGGGTAGGATGATT-3'	110	(25)

Table 3. Evaluation of essential oils at different doses for the inhibition of methane production and effect on digestibility in wheat straw based diet (Roughage 50: concentrate 50).

Treatment combinations	CH₄ (ml/g of substrate)	CH₄ (mM/g DM)	CH₄ (mM/g DM)	% IVDMD
Control Diet(50:50)	30.24	1.18	0.20	58.37
C+T1 (50 mg/L)	38.17	1.49	0.25	59.89
C+T1 (150 mg/L)	34.56	1.35	0.21	64.72
C+T1 (450 mg/L)	25.67	1.00	0.19	51.64
C+T2 (50 mg/L)	41.07	1.60	0.24	67.25
C+T2 (150 mg/L)	31.04	1.21	0.19	63.78
C+T2 (450 mg/L)	30.63	1.20	0.20	59.67
C+T3 (50 mg/L)	33.90	1.32	0.21	64.02
C+T3 (150 mg/L)	28.87	1.13	0.17	66.99
C+T3 (450 mg/L)	9.41	0.37	0.08	43.52
C+T4 (50 mg/L)	36.97	1.44	0.23	64.23
C+T4 (150 mg/L)	35.57	1.39	0.23	44.89
C+T4 (450 mg/L)	5.28	0.21	0.04	50.31
C+T5 (50 mg/L)	35.60	1.39	0.21	66.80
C+T5 (150 mg/L)	41.83	1.63	0.26	62.65
C+T5 (450 mg/L)	31.47	1.23	0.20	62.08
CD	14.66	0.57	0.09	12.20

T1: Cinnamon leaf oil; T2: Clove bud oil; T3: Thyme oil; T4: Origanum oil; T5: Rosemary oil
CD: values were significant at 5 percent

Table 4. *In vitro* effects of origanum and thyme oils on the digestibility and methane production in wheat straw based high, medium and low fibre diets.

Diet	Treatment combinations	Parameters					
		IVDMD (%)	PF	MBM (mg)	CH₄ (ml/g DM)	CH₄ (mM/g DM)	DDM (mM/g DDM)
HFD	Control	76.19	3.79	61.81	23.46	0.92	0.12
	Origanum oil (450 mg/L)	74.78	6.16	94.99	14.42	0.56	0.07
	Thyme oil (450 mg/L)	56.87	6.80	73.62	12.13	0.48	0.08
MFD	Control	77.07	3.80	62.84	21.70	0.85	0.11
	Origanum oil (450 mg/l)	60.66	5.66	72.39	14.59	0.57	0.09
	Thyme oil (450 mg/L)	57.57	5.27	66.02	18.38	0.72	0.12
LFD	Control	70.58	3.40	47.59	24.22	0.95	0.13
	Origanum oil (450 mg/L)	32.77	5.53	37.60	6.67	0.26	0.08
	Thyme oil (450 mg/L)	58.29	14.70	98.58	4.46	0.17	0.03
SEM	Diet (D)	8.92	1.78	N.S.	1.91	0.076	0.014
	Treatment (T)	8.92	1.78	N.S.	1.91	0.076	0.014
	D x T	15.46	3.08	35.00	3.31	0.13	0.02

Table 5. Effect of origanum and thyme oils on the volatile fatty acids and ammonia nitrogen in wheat based high, medium and low fibre diets

Diet	Treatment Combinations	Parameters				
		TVFA (mM/100 ml)	Acetate (%)	Propionate (%)	Butyrate (%)	NH₃-N (mg/100 ml)
HFD	Control	12.00	80.55	14.70	4.75	39.20
	Origanum oil (450 mg/L)	10.83	78.69	7.95	13.36	28.47
	Thyme oil (450 mg/L)	6.83	78.71	12.33	8.96	15.40
MFD	Control	8.17	77.11	17.67	5.22	21.47
	Origanum oil (450 mg/L)	9.08	81.71	8.26	10.03	25.20
	Thyme oil (450 mg/L)	7.00	85.75	8.66	5.59	20.07
LFD	Control	10.33	79.95	15.93	4.13	20.53
	Origanum oil (450 mg/L)	7.00	81.91	9.86	8.23	21.47
	Thyme oil (450 mg/L)	7.83	84.71	10.08	5.21	14.00
SEM	Diet (D)	1.21	N.S	N.S	1.71	2.43
	Treatment (T)	1.21	2.65	2.38	1.71	2.43
	D x T	2.09	4.58	N.S	N.S	4.21

CD: values were significant at 5% during *in vitro* trials

Tables 6. Quantitative real-time PCR quantification of different rumen microbial groups in control as well as essential oil supplemented in vitro rumen fermentation reactions.

Groups	Total bacteria ($\times 10^{10}$)	Methanogens ($\times 10^7$)	Protozoa ($\times 10^7$)	Fungi ($\times 10^7$)
Control	2.12 $\pm$ 0.008	5.9 $\pm$ 0.01	1.0 $\pm$ 0.01	9.6 $\pm$ 0.1
Origanum oil	2.71 $\pm$ 0.02	6.7 $\pm$ 0.01	1.3 $\pm$ 0.03	8.9 $\pm$ 0.07
Thyme oil	1.98 $\pm$ 0.007	5.6 $\pm$ 0.08	1.3 $\pm$ 0.01	7.8 $\pm$ 0.1
CD	0.00	0.03	0.08	0.56
<i>p</i>-value	0.09	0.00	0.00	0.00